Counts, laws and denominators

Fix report g and information y.
Weights are nonnegative and sum to one.
ν = s (supplied) or P (evaluation).

$$M = \binom{n}{2} > 0, \quad x_{ij} = \mathbf{1}\{g_i < g_j\}$$

$$N(g) = \sum_{i \neq j} x_{ij}$$

Raw disclosure N/M is fixed by g.

Weak error (≤)

$$D_k(g, \Theta) =$$

$$\sum_{i \neq j} x_{ij} \mathbf{1}\{\theta_{ik} \leq \theta_{jk}\}$$

Average under ν at fixed y:

$$E_k^{\nu}(g) = \mathbb{E}_{\nu}[D_k \mid y]$$

$$E_k^{\nu}(g)/N(g), \quad N(g) > 0$$

Strict reversal (<)

$$\sum_{i \neq j} x_{ij} \mathbf{1}\{\theta_{ik} < \theta_{jk}\}$$

Average under ν at fixed y:

$E_k^{b,\nu}(g)$ expected strict-error count

$$E_k^{b,\nu}(g)/N(g), \quad N(g) > 0$$

$$E_k^{b,\nu}(g)/Q_k^{\nu}(g), \quad Q_k^{\nu}(g) > 0$$

Distinct opportunity (≠)

$$\sum_{i \neq j} x_{ij} \mathbf{1}\{\theta_{ik} \neq \theta_{jk}\}$$

Average under ν at fixed y:

$Q_k^{\nu}(g)$ expected distinct mass

$$Q_w^{\nu}(g)/T_w^{\nu}, \quad T_w^{\nu} > 0$$

Weighting and total opportunity

$$Q_w^{\nu}(g) = \sum_k w_k Q_k^{\nu}(g)$$

$$T_w^{\nu} = \sum_{i<j} \sum_k w_k d_{ij,k}^{\nu}$$

$$d_{ij,k}^{\nu} = \Pr_{\nu}(\theta_{ik} \neq \theta_{jk} \mid y)$$

T: all original pairs; Q: disclosed directions.

Inspected tie_aware_v1 field names

API D ↔ $Q_w^s(g)$; API T ↔ T_w^s

Supplied-law aliases; book D_k remains
a realized weak wrong-count.

Zero-denominator and varying-report
qualifications follow in the text.